



Fairness of AI in Predicting the Risk of Recidivism: Review and Phase Mapping of AI Fairness Techniques

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ABSTRACT

Artificial Intelligence (AI) is applied in almost every public sector because of its positive impacts. However, AI's ethical aspects and trustworthiness constitute a significant uproar and concern among different AI stakeholders due to AI's adverse effect on users when the AI system lacks cautionary measures. AI is used in the criminal justice system for predicting recidivism risk. However, AI's negative impact translates into bias and high incarceration towards a group of defendants in a population assessed for recidivism risk. This paper focuses on fairness as a requirement of a trustworthy AI framework previously proposed to ascertain the appropriate application of AI systems in predicting recidivism. This paper aims to raise awareness about the fairness of AI models and stimulate further research and deployment of efficient and effective exploitation of fair and trustworthy AI models in the criminal justice system when predicting recidivism. Fairness has been a significant concern for criminal justice system stakeholders and has received considerable attention with more theoretical and practical studies than other trustworthy AI requirements. Hence, this paper reviews state-of-the-art fairness, outlines valuable findings, and proposes future directions to achieve fair AI systems for predicting recidivism risk. In addition, this paper ensures mapping existing technical works in the literature to the fairness pipeline corresponding to the criminal justice system's AI development phases.

CCS CONCEPTS

• **Trustworthy Artificial Intelligence** → Fairness; • **Criminal Justice System** → Recidivism;

KEYWORDS

Trustworthy Artificial Intelligence, Criminal Justice System, Fairness, Trust, Recidivism

ACM Reference Format:

Michael Mayowa Farayola, Irina Tal, Takfarinas Saber, Regina Connolly, and Malika Bendeckache. 2023. Fairness of AI in Predicting the Risk of Recidivism: Review and Phase Mapping of AI Fairness Techniques. In *The 18th International Conference on Availability, Reliability and Security (ARES 2023)*, August 29–September 01, 2023, Benevento, Italy. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3600160.3605033>

1 INTRODUCTION

Recidivism is the rearrest, re-conviction and cycling back of an offender into the criminal justice system after being released on parole or probation [29, 52, 53, 59, 60]. Different statistics demonstrate that recidivism is a serious problem in our society. For instance, a study on the rate of imprisonment and recidivism among prisoners released by the Irish prison service [13], based on re-offending and re-conviction data in 2013, indicated that the recidivism rate of prisoners released within three years was at a high rate of 63.3%. However, more than 80% of released prisoners re-committed a crime within 12 months of release. The population studied was 7,701 offenders. The study also looked into the rate of recidivism relating to the age and gender of offenders. However, the population was male-dominated, with a percentage of 92.5%. The study further shows that males have a higher recidivism rate than their female counterparts, with a percentage of 63% for males and 57% for females. The female percentage indicates that more than half of the females recidivate. We found a similar pattern of this kind of recidivism rate in [40, 41]. All these statistics serve as measures to show the dire concern of recidivism risk to the public.

In recent years, many criminal justice system stakeholders have focused on the impact of Artificial Intelligence (AI) applications in predicting the risk of recidivism. AI's positive effects outweigh its negative implications, such as its ability to bring efficiency in predicting convicted individuals' likelihood of recommitting an offence [4, 11, 22, 23, 49, 58]. However, the negative impact of AI is



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ARES 2023, August 29–September 01, 2023, Benevento, Italy
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ACM ISBN 979-8-4007-0772-8/23/08.
<https://doi.org/10.1145/3600160.3605033>

a stumbling block to AI models' acceptability in predicting recidivism risk. These include instances of bias against a set of convicted individuals or lack of accountability when the decision made by AI systems goes wrong. Therefore, several researchers have implemented AI development methods to reduce AI's negative impacts on the offenders assessed for recidivism risk while increasing its positive impacts. These negative concerns prompted us to proffer a lasting solution to end AI technologies' unacceptability in predicting recidivism risk.

In 2019, European Commission (EC) published the Ethics Guidelines for Trustworthy AI [1] that focused on addressing: (1) how best AI users can trust the processes of AI applications to actuate AI acceptability and dependency in the public sector and (2) how the AI system itself will be trustworthy to deliver the desired results to the stakeholders. In establishing a trustworthy AI, the EC proposed non-exhaustive requirements to ascertain stakeholders' acceptability, dependency, and trust in the AI model application processes [1, 12]. These requirements range from Human agency & Oversight, Technical Robustness & Safety, Privacy & Data Governance, Transparency, Diversity & Fairness, Societal & Environmental Well-being, and Accountability.

This paper focuses on fairness as a requirement of trustworthy AI when predicting the risk of recidivism. Fairness has received theoretical and practical studies, whereas other reliable AI requirements are more or less on a theoretical level. Hence, the paper examines the state-of-the-art practical development of fair AI when predicting recidivism. Through this, we propose future directions to ascertain further fair AI systems in predicting recidivism in the criminal justice system.

This paper is structured as follows: Section 2 discusses definitions of fairness and its sources. Section 3 analyzes the state-of-the-art on fairness and outlines future research directions. Section 4 concludes the paper.

2 FAIRNESS

There is an imperative need for AI models to be fair with their increasing application in different jurisdictions around the world in predicting the risk of recidivism. How to achieve fairness is an issue due to the automation of bias, the inability of AI systems to create unbiased conclusions, the maintenance of bias, and the amplification of previous discriminations [19, 55]. There is an ongoing controversy and debate on a standard definition of fairness that will conform to all instances or at what AI development stage fairness should be evaluated for predicting recidivism [49]. A study on the pursuit of fair and accurate Machine Learning (ML) for criminal recidivism prediction [58] quotes that emerging consensus on a precise definition of fairness must rely on AI model characteristics and domain-specific expertise. The following section briefly mentions standard definitions of fairness proposed in the literature applicable to recidivism risk prediction.

2.1 Definitions of fairness in the literature

A review on discrimination, bias, fairness, and trustworthy AI [55] gave a technical point of view of fairness as measures taken to improve models without changing or manipulating them for reasons unrelated to the interests of users and also ensuring the absence of prejudice towards individuals or groups. Several definitions of

fairness have been proposed and distinguished in the literature under two main categories: 1) Individual fairness and 2) Group fairness [7, 14, 21, 32, 35, 37, 51, 57, 63]. Individual fairness ensures that similar individuals are attributed the same outcomes and treatments. Group fairness ensures that individuals across all groups receive the same treatment. Some definitions formulated are fairness through awareness, fairness through unawareness, equal opportunity, counterfactual fairness, preference-based fairness and equalised odds. Other are conditional statistical parity, demographic parity, calibration, predictive Parity, predictive Equality, and treatment equality. For interested readers, these fairness definitions are discussed extensively in [6, 20, 21, 35, 51, 56].

2.2 Sources of unfairness

Bias and discrimination are two terms leading to unfairness in AI applications for predicting recidivism. Many scholars have attributed the primary source of bias or discrimination in AI systems to the dataset used in training the AI models [7, 8, 14, 19, 24, 45, 53]. In addition, it is evident that there is a presence of unfairness in model development called model bias and the evaluation processes of the AI system called evaluation bias [32]. However, it is imperative to understand the difference between discrimination and bias. There is always an interchangeable use of these two terminologies [18, 36, 55].

Discrimination is a source of unfairness that results from the unequal treatment of an individual or group directly or indirectly linked with protected attributes (such as race, gender, ethnicity, etc.) [7, 18, 36, 55]. Discrimination divides into two: direct discrimination and indirect discrimination [36, 55, 62]. Direct discrimination can also be referred to as *disparate treatment* or *intended discrimination* [16]. Indirect discrimination is referred to as *disparate impact* or *unintended discrimination* and which in itself is not illegal [16]. In the context of the judiciary system, direct discrimination is the scenario whereby an offender receives unfair treatment because the offender is a group representative. An explanation for indirect discrimination will be a scenario whereby a decision is made, through unprotected attributes like zip code, by implementing high policing in a community because of the past high crime rate in such a community. This is ideal for safeguarding the environment, but this will lead to more arrests in the community and obliquely prompt racial discrimination. A study proposed algorithms for discovering and removing direct and indirect discrimination. Interested readers can look at [62] for a more in-depth understanding.

In addition, the categories of discrimination are explained: Explainable and In-explainable discrimination [36, 55]. Explainable discrimination is a discriminatory output by a predicting algorithm that can be explained and justified by features influencing the decision. In-explainable discrimination occurs when the discriminating output of predicting algorithmic outcomes lacks a convincing explanation. In all, discrimination has been a concern when building predictive algorithms, and understanding at what stage discrimination leads to bias or unfairness, is still a bottleneck.

Bias is a source of unfairness that results from potentially detrimental or unexpected characteristics of the data that cause algorithmic outputs to deviate from actual value [15, 19, 24, 42, 54]. Bias is the decisions made based on preconceived ideas or preconceptions rather than an unbiased assessment of the facts [26]. The

implementation of corporate objectives and emphasis on creating AI applications fit for such purposes is mostly the starting point of bias in AI development [36, 50]. Nevertheless, most instances of these objectives implementations lead to unintentional biases. Every growing organisation uses AI systems to reform and optimise its services to its customers. Suppose it lacks the right processes, accidental behaviour or bias may appear and endanger the creation of such an AI system, jeopardising the organisation's reputation and dwindling the customers' trust. According to the literature on bias, the primary source of bias in AI development is the dataset, primarily by using historical data ascribed with human prejudice activities when collecting the data. Hence, AI algorithms significantly tend to pick up on human bias. There are three significant types of bias in the AI lifecycle: 1) Data Bias, 2) Model Bias and 3) Evaluation Bias. In these types of bias, other forms emanated, further explained alongside the significant kinds of bias in subsequent sections.

2.2.1 Data bias. Most datasets used to train AI algorithms are from historical datasets, mostly embedded with human prejudice during their collection and a deficit of proper mechanisms to ascertain a fair dataset. The use of historical datasets can lead to measurement bias which is the case of over-prioritising a feature because the feature has been historically used in making decisions [19, 44, 45, 55]. Another source of data bias is in implementing organisation goals into AI development. The organisation wants to use the AI system and what sample distribution or features will be selected to train the AI model. Likewise, how does the organisation intend to represent the selected features numerically? When the sample selection does not accurately reflect reality, it is term sample bias. Sample bias occurs when there is a lack of even sample size distribution across groups. Creating a training dataset for AI models is often the most tedious task in developing an AI system, and proper mechanisms should be in-placed to ensure fair training datasets. Another factor that triggers data bias is the problem of mismatched datasets. A mismatched dataset is an instance whereby the test data differs from the training data. When a mismatch of datasets occurs, it facilitates the poor performance of AI models. The issue of training dataset manipulation or unnecessary correlations between the input attributes used to train the model and the model output are additional causes of data bias. [50].

2.2.2 Model Bias. This occurs when the algorithm introduces bias in AI development even if we have a fair dataset [36, 44]. Likewise, a trigger of model bias is the AI algorithm prioritisation of some input features over another, leading to bias without justification [19, 32]. AI algorithms' internal workings learn by recognising and spotting patterns in the dataset [55]. Most of the time, it is impossible to determine which features AI models train on. As a result, additional efforts should look at the internal working of AI models during the training process. The adjustment of AI algorithm parameters is another factor that causes model bias. The hyper-parameters developers defined at the AI algorithm development phase are the basis on which AI models learn to make decisions. Developers should use caution and be articulate when selecting hyper-parameters to reduce bias further. "time" bias is another factor that could contribute to model bias [55]. The passage of time may impact how well AI systems perform as training features or attributes change over time.

2.2.3 Model Evaluation Bias. This occurs when the AI model's performance is assessed with the wrong evaluation metrics. Another trigger of evaluation bias is biased feedback from AI users. Often, organisations tend to re-model the AI system based on the feedback received from users [32]. The feedback collection can be during the AI model's testing phase or after the AI system's deployment. The kind of bias that evolves due to the interaction of users with the deployment AI system and receiving feedback to re-design the AI system is emergent bias [55]. Hence, it is imperative for organisations to thoroughly examine the feedback to ensure unbiased before re-implementing feedback into the AI model. Choosing the proper assessment metrics and carefully spotting different kinds of prejudice from user feedback is critical to avoid evaluation bias.

In conclusion, there has been controversy on the inter-relatedness of these two sources of unfairness: bias and discrimination. Some scholars think that discrimination leads to bias and vice versa in developing and deploying AI models. Nevertheless, there is still a need for a precise understanding of how algorithmic prejudice and discrimination are related [18].

2.3 Fairness pipeline

A fairness pipeline considers mitigating or removing discrimination and bias in developing AI systems to facilitate fair AI. Therefore, this section discusses the techniques proposed to ensure a fair AI system. The fairness pipeline has three phases that will be detailed next: pre-processing, in-processing, and post-processing [6, 14, 31, 42, 43, 57, 63, 64].

2.3.1 Pre-processing phase. This is the first phase used to guarantee the ethical growth of an AI system [17, 32]. From the data collection until the creation of the final training dataset, this method focuses on processing the dataset to develop a fair AI model. By precisely identifying and preparing datasets, researchers may report the faults detected in their dataset [17]. The quality and unbiasedness of the training dataset primarily determine the AI model's fair prediction.

2.3.2 In-processing phase. This is the second phase of guaranteeing the ethics and trustworthiness of the AI system [57]. The method focuses on the tuning of AI algorithms and the development of AI models. The hyper-parameters used in training AI algorithms are carefully selected and conform to the application domain. AI developers should opt for AI algorithms less prone to bias and ensure the AI model follows due hyper-parameter tuning processes.

2.3.3 Post-processing phase. The third stage of assuring fairness in AI systems is post-processing methods. They are aimed at the assessment stage of the AI lifecycle to address concerns of prejudice and bias that may arise. Post-processing in algorithmic fairness is a versatile technique for reducing bias from machine learning systems currently used in production [46]. Post-processing's main advantage is that it eliminates the need for costly retraining.

The following section looks into the state-of-the-art methods used in each phase that characterizes a fairness pipeline to address the issue of unfairness in AI models for predicting the risk of recidivism.

3 PRACTICAL TECHNIQUES USED FOR ACHIEVING FAIR AI IN PREDICTING THE RISK OF RECIDIVISM.

Fairness is the most studied trustworthy AI requirement when applying AI models to predict the risk of recidivism. This section enumerates and discusses the state-of-the-art practical techniques for actualizing fair AI models for recidivism at the different AI development phases (see table 1).

As mentioned in section 2.3, three traditional fairness phases are established in the literature to actualize fair AI models for recidivism risk prediction. These are the pre-, in-, and post-processing phases. This paper maps the different state-of-the-art methodologies to these phases and discusses them in detail. Figure 1 presents an overview of the papers analysed from the literature concerning their focus on the three phases. Note that some papers address unfairness in two fairness pipeline phases; we map these to the prioritised phase. A summary of all the literature discussed is in Table 1.

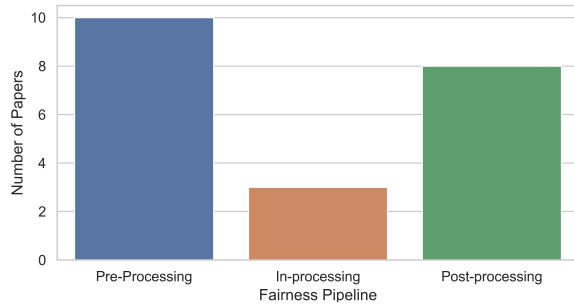


Figure 1: Overview mapping: Analysed papers and their focus on the 3 phases of the fairness pipeline

3.1 Pre-Processing Phase

Comprehensive research on bias and accuracy in predicting prisoner recidivism [29] built novel singular race models by dividing recidivism dataset based on offenders' race. This approach is motivated by standard practices prioritising removing racial information from recidivism datasets to mitigate racial bias [14, 29]. The design of the singular race model is to train and test one sub-population at a time without explicitly including race information. Compared with a base model developed using data from all races, the base models predict recidivism with a high FPR and low FNR for African American races over other races. The base model put African Americans on the verge of high incarceration over other races. However, the singular race model did not produce much better results as it further exacerbated the base model's bias but produced a high accuracy. The authors [29] assert that this is because the dataset used is unbalanced and has limited features. In addition, despite the singular race models built on precise race information, the study questioned which features strongly correlate with race data to have increased the predictive bias. In conclusion, mitigating bias through the singular race approach must be explored further as it is promising to address racial discrimination in recidivism datasets. A future direction of mitigating unfairness in the singular race approach

will be to add more features expressing personal information about the offenders than just the demographics and investigate possible race-related features.

Bhanu et al. [27] carried out a research work introducing a novel fairness measure, Bias Parity (BP) Score, and a feature-rich representation derived from the temporal aspects of the dataset. Bias Parity facilitates the selection of the least biased models out of a set of models trained on the various iterations of augmented feature sets. The proposed fairness metric is motivated by the lack of a quantitative measure of bias for the predictive model usable for any statistical measure [27]. Bhanu et al. [27] use an offender's history to mitigate bias and increase accuracy as it invalidates the conventional notion of accuracy and fairness tradeoffs. The BP score permits model comparison using a single numerical number based on the bias embedded in the model. This work forecasts recidivism among parolees using data from the "Recidivism of Prisoners Released in 1994" investigation. The study used reconviction to generate several models and set them as an indicator of recidivism. This approach contradicts the standard method of using arrest as an indicator of recidivism. The author evaluated for bias by testing with all races (Caucasian and African American) and twice with individual races while removing race from the test data and using the BP score of a range of fairness metrics. However, a computational cost was associated with training and testing numerous models with varied history lengths to select the least biased model, even if they received compensation for better performance and less bias. A limitation of this approach is the use of reconviction against using rearrests as an indicator of recidivism. However, if there is human bias when making the reconviction decisions, the algorithm will embody them.

A study on data pre-processing for discrimination prevention [14] proposed a novel probabilistic formulation of data pre-processing for reducing discrimination. The author introduced this novel approach to control group discrimination, limit distortion in individual data samples, and maintain utility. The approach aims to stop the unfair treatment of offenders because of their affiliation with a protected group. This approach is applied to the COMPAS recidivism dataset to illustrate the technological viability of reducing the disparate impact of rearrests on various demographic groups while maintaining utility and individual fairness. The claim is that removing protected attributes from the training dataset is insufficient to solve indirect discrimination. This claim is also evident in [8, 29]. The novel data pre-processing approach is made such that given an original dataset (D, X, Y) where D is the protected attributes, X is non-protected attributes, and Y is a random outcome variable, X and Y are transformed through a probabilistic method to satisfy discrimination control (group fairness), distortion (individual fairness) and utility preservation constraints while maintaining the protected features to achieve a new dataset [14]. At the level of probability distributions, utility and discrimination are defined. Distortion is managed on a per-sample basis to limit the impact of the transformation on specific offenders and guarantee individual fairness. Flavio et al. [14] observed that its pre-processing method reduces bias, particularly when utilising the original dataset with or without eliminating protected features. In its experimental results, the empirical discrimination is decreased with a cost on accuracy when the novel pre-processing approach is used compared to when it is not. A detailed explanation is in [14].

Many conventional techniques emphasize achieving the value of disparate impact greater than 0.8 (80% rule), that is, eradicating bias related to protected features while ignoring the problem of intersectional discrimination. Furthermore, on pre-processing techniques, a work on intersectional bias [33] proposed a One-vs.-One mitigation strategy to address the disregard for unjust treatment of members of subgroups of protected groups. This approach is motivated by the fact that no research study has addressed the unfair situation of offenders belonging to a subset of a protected group when considering multiple sensitive attributes. Kenji et al. [33] applied the proposed mitigation strategy to the COMPAS dataset, which has systemic gender bias correlated to intersectional discrimination [25, 33]. The proposed approach considered the gender columns in the COMPAS dataset by dividing the columns into four subgroups (i.e. (female, non-white), (female, white), (male, non-white), (male, white)) and further paired to have six sub-datasets containing all possible subgroup pairs. Compared with conventional mitigation methods of removing bias, the One-vs.-One mitigation technique helps mitigate intersectional discrimination. Its experimental result shows that the technique reduced the disparity between subgroups more than the traditional ways in all cases with no significant decline in accuracy. In addition, this technique helps to improve the following fairness criteria; demographic parity, equalized odds, and equal opportunity.

In contrast to several studies on mitigating unfairness in recidivism datasets, particularly racial inequalities, a study on detecting racial inequalities in the criminal justice system [10] proposed using mugshots to generate racial categories and to fill in missing race data. This approach is motivated by the perpetual loss of racial data in training data. Hence, the study focuses on understanding how AI models identify racial class and how the best deep learning model can fill in missing race information. Also, the study argues that instead of the common approaches of addressing missing datasets, as they can affect the prediction of AI models and exacerbate unfairness [10, 19], mugshots can be incorporated through the use of facial processing technology.

A study evaluating causes of bias [38] took a different direction to identify two data biases (concerning women and foreigners) that lead to possible causes of algorithmic bias. The study explored two pre-processing mitigation methods; stratified oversampling algorithm to equalize the base rate and removing the impact of protected features. The dataset used for the analysis is the Catalonia (Spain) juvenile justice system because most literature addressing unfairness in the criminal justice system is US-based. The study analyzed the differences in the prevalence of re-offences among protected groups and the influence of protected groups on prediction, both identified as data bias resulting in algorithmic bias. Statics features are demographic characteristics and past criminal history, while current substance abuse, peer rejection, or hostile behaviour are examples of dynamic features. The authors assumed that static features correlate more with protected features and induce more disparity between group fairness metrics than the dynamic features. Furthermore, a thing to note is that algorithmic models that depend on static features exhibit less group fairness. As much as the two pre-processing methods can increase predictive parity, they do not ascertain group fairness as both are deficient in satisfying other group fairness[38]. In summary, the criminal justice system using

AI models relying solely on static features can be coercive rather than transforming.

3.2 In-Processing Phase

In-processing phase is a crucial aspect of mitigating unfairness in AI algorithms [57]. A study on algorithmic decision-making and the cost of fairness [9] formulated algorithmic fairness as constrained and unconstrained optimization algorithms. Building constrained optimization algorithms require applying multiple and race-specific thresholds to achieve defendants' risk scores, such as detaining a black defendant with a risk score above 5 and detaining a white defendant with a risk score above 7. In contrast, unconstrained optimization algorithms require using a single and uniform threshold for all defendants, irrespective of race. The formal aims to maximize public safety, while the latter aims to ensure equality, reducing race disparities. However, there are trade-off concerns about which of these two forms of algorithmic fairness to use when predicting the risk of recidivism, as there is tension about whether to focus on mitigating racial disparities or maximizing public safety. This study [9] helps to foster discussions for policymakers and researchers on the design and application of AI models for predicting the risk of recidivism. Experimenting on the Broward County dataset (COMPAS), the study showed that racial inequalities result from public safety optimization; in contrast, releasing more high-risk criminals to fulfil fairness criteria would harm public safety. As a result, the research recommended that high-risk offenders participate in community monitoring programs.

In addition, a study on penalizing unfairness in binary classification [3] expressed concern about achieving fairness in classification algorithms, as very few techniques address possible ways of attaining algorithmic fairness. The approach adopted in the study is motivated by algorithmic regularization. Regularization is a function of learned hypothesis, which address model complexity and helps to reduce overfitting. The study focuses its proposed new penalty or regularization terms on achieving similar false positive (FPR) and false negative rates (FNR) for subgroups in the populations. Two penalizers were defined to achieve the study research goal of obtaining similar FPR and FNR: Absolute Value Difference and Squared Difference, both hypothesis-dependent and data-dependent focusing on boundary-based classifiers. Applying these two penalizers on logistic regression using the COMPAS dataset for training and testing, the study conducted its experiment and compared its methodology's performance to three previously developed approaches. In the result and conclusion, the authors found that these two penalizers achieve similar accuracy with previous methodologies in the literature but improve with significantly better fairness for defendants than previous methodologies. In addition, the authors proposed that to ensure FNR- and FPR-matching, it is advisable to initially apply unfairness mitigation methodologies at the in-processing phase as against the post-processing phase to avoid sub-optimal classifiers. The study found out that post-processing unfairness mitigation techniques may often lead to reduced accuracy in favour of some defendants over another [3].

Due to less concentration on addressing unfairness in regression settings (where target values are continuous) as much significant research focus on classification settings, a study on a convex framework for fair regression [5] introduced a family of fairness

regularizers for regression problems. The study presented a family of fairness metrics for regression models that take the form of a fairness regularizer and apply them to the standard loss functions for linear and logistic regression. The author formulated the fairness metrics to cover the spectrum of group, intermediate, and individual fairness. In addition, the framework developed for this research follows the approach of removing protected attributes in the training and testing data used for developing their model. The study also established a data-dependent measure termed the Price of Fairness (PoF), which standardizes the fairness loss across data sets to lie on the same scale [5]. The PoF is calculated by determining the "Pareto curve" or employing different weights on the fairness regularizer. The study applied its approach to the COMPAS datasets. It concluded that the frequent occurrence of accuracy and fairness trade-offs in several literatures depends on fairness regularizer choice and the dataset used. Hence, this is suggestive that expertise must take absolute caution when defining fairness for any given AI model used when predicting recidivism risk [5].

3.3 Post-Processing Phase

A study on the problem of prior probability shifts carried out in [8] claimed that there is no known method for ensuring Proportional Equality (PE) fairness. PE accounts for prior probability shifts whereby the training set distribution differs from the test set [8]. Neglecting distributional changes causes predictive classifiers to be biased towards the population subgroups (Male, Female). This stated problem occurs in the COMPAS recidivism dataset used for predicting recidivism. Therefore, Arpita et al. [8] proposed a Combinatorial Algorithm for Proportional Equality (CAPE), which utilizes quantification techniques (a post-processing approach), sampling, and an ensemble of classifiers to ensure PE-fair predictions. CAPE aims to minimize the prevalence difference (PD) metric to ensure PE fairness. The quantification technique is a scaled probability average (SPA) to estimate true and false positive rates.

In the experimentation and evaluation process, using the COMPAS dataset, the author [8] compared the CAPE with an accuracy-maximizing classifier (Max_Acc) and other (in-, pre-, and post-processing) fair algorithms implemented in IBM AI Fairness 360 toolkit such as Reweighting, Adversarial Debiasing, variants of Meta-fair, Calibrated Equalized Odds Post-processing and Reject Option Classification. These algorithms are evaluated based on the tendency to achieve PE fairness and performance compared to some metrics such as FNR-diff, FPR-diff, Accuracy-diff, and PD. In the result, Arpita et al. [8] validated the presence of prior probability shift in COMPAS by the significant differences in the true prevalence of the training and test datasets. In addition, CAPE outperforms every algorithm while exhibiting equalized odds. Arpita et al. [8] further stated that they expected to attain proportional equality for any classifier satisfying statistical parity. In all, CAPE has shown to be robust to perform well on several metrics, and a further extension of CAPE will be to explore on *concept drifts* distributional change.

Predictive parity and group calibration are sufficiency-based measures metrics often used to evaluate recidivism prediction instruments [39, 61]. Emphasises are that *observational* fairness criteria are unfitting for risk assessment instruments (RAIs) for recidivism prediction [39]. A study on fair Bayes-optimal classifiers under predictive parity [61] researched sufficiency-based measures,

significantly predictive parity. The author stated that in selective classification(which is helpful in decision-making), sufficiency-based representation learning leads to fairness [34, 61]. The study led to the developing of the FairBayes-DPP (Difference in Predictive Parities) algorithm aimed at predictive parity and tested on the COMPAS dataset. In their research, Zeng et al. proved that all fair Bayes-optimal classifiers under predictive parity are GWTRs (group-wise thresholding rules).

More on techniques addressing unfairness in post-processing is a research work in [39] focused on the fairness criterion, counterfactual equalized odds. Counterfactual equalized odds utilize possible outcomes that allow users to negotiate the tradeoff between fairness and performance. The study prioritizes equalized odds over predictive parity, arguing that predictors that satisfy predictive parity are likely to exhibit disparate impact. Alan et al. [39] concluded, using the COMPAS dataset, that approximate counterfactual equalized odds is more feasible to mitigate disparate discriminatory impact than any existing fairness criterion. However, this assertion by Alan et al. [39] contradicts the arguments made in [61].

Setting up AI models to predict recidivism ethically and with legal constraints will significantly reform the criminal justice system and build stakeholders' trust in the predictive tools. With these in effect, a study on recycling privileged learning and distribution matching for fairness [48] came about a unified ML framework capable of handling different fairness metrics, their combinations, and future definitions of fairness. Privileged learning is a case of features available during learning but absent during testing, and distribution matching is a phenomenon of having two different datasets behaving in similar ways [48]. The study focuses on four definitions of fairness: demographic parity, fairness through unawareness, equality of opportunity, and equalized odds. The author achieved the unified ML framework by recycling privileged learning and distribution, matching and harmonizing them to meet diverse definitions of fairness [48]. Most of the defined fairness could help predict offenders' risk of recidivism and address the different tradeoffs that might occur during the evaluation of recidivism risk assessment systems.

To address equalized odds tradeoff with some fairness criteria, a study on conditional learning of fair representations [65] proposed a novel algorithm called CFAIR (for conditional fair representations) to mitigate and achieve equalized odds and accuracy parity. The algorithm's formulation is based on a balanced error rate (BER) over the protected characteristics and the target variable. Experimenting using the COMPAS dataset, BER is proven to achieve equalized odds and accuracy parity simultaneously and CFAIR is emphasized as more appropriate to use on a balanced dataset. In addition, the study used the conditional alignment of the representations over marginal distributions.

A study on using algorithms to address fairness tradeoffs inherent in predicting recidivism [53] has pointed out the importance of not blinding recidivism risk algorithms to sensitive attributes, as blinding the algorithms affects the evaluation procedure of the algorithm. The author claimed that not blinding recidivism risk algorithms approach improves calibration and positive predictive values and reduces imbalanced error rates [49, 53]. Skeem et al. [53] showed in the study how three different "debiasing" algorithm methodologies (Proxy eliminated algorithm, race eliminated algorithm, and criminal history discount algorithm) influence fairness

tradeoffs. The fairness-trade-offs examined are calibration against the positive predictive value and false positive against false negative error rates. In their approach, being rearrested for a violent crime is set as an indicator of recidivism. In conjunction, Skeem et al. [53] corroborated the method used in [27] on using offender's history to predict recidivism but suggested it could be demeaning by the prejudiced approach in policing and decision-making about arrests which could affect a particular sub-population.

A study [28] proposed that using criminals' history will effectively improve recidivism risk assessment tools' accuracy and further help mitigates bias. Jain et al. [28] developed its model using fairness through the unawareness (race-blind dataset) approach. The training data (Recidivism of Prisoners Released in 1994) comprises the criminals' current and past history of arrest cycles. Likewise, arrest and reconviction are indicators of recidivism. The selected model is chosen based on exhibiting a reduced bias for several sub-populations with increased accuracy through FPR parity. The developed model was used to predict all crimes and sexual crime recidivism. The study concluded that criminal history increases model accuracy while reducing bias.

Another research [47] looks at how calibrated probability estimates can be maintained while lowering error disparities across various population groups. The approach drives the necessity to preserve risk instruments calibration when utilized in practice and the inherent tradeoff with other fairness properties. [47]. Geoff et al. [47] further noted that using risk tools calibrated to groups defined by race is advisable to avoid risk tools having unpredictable and potentially undesired results. The calibrated risk tools will stop encouraging judges to consider race when making decisions. However, Pleiss et al. developed the calibrated model such that Equalized Odds-Derived is applied as the post-processing method on the COMPAS dataset and uses the Equalized Odds-Trained techniques to build classifiers directly encode the Equalized Odds constraints. This approach uses existing calibrated classifiers and post-processes them to obtain calibrated Equalized Odds relaxation. However, the limitation of Pleiss et al.'s work was that calibration is incompatible with error-rate limitations. Nevertheless, if criminal justice practitioners use recidivism predictive tools calibrated to race, there will be less discrimination in the form of FP and FN rates. Lastly, a lower-error solution is reachable if developers consider calibration in developing recidivism prediction tools.

A study on enhancing the Recidivism Prediction tool with Machine Learning [30] adopted the method in [47]. It is a known fact that COMPAS is the most studied dataset for predictive fairness [3]. Karimi et al. applied the adopted method in [47] to a new dataset called RisCanvi dataset with three different ML models built (Logistic Regression (LR), Multi-layer Perceptron (MLP), and Support Vector Machines (SVM)) for violent and general recidivism prediction. Compared with the RisCanvi protocol (a statistical way of assessing offenders' risk of recidivism), LR produced the best result for predicting general and violent recidivism. Point to note: the developed models were calibrated to nationality and age sub-groups. The calibration led to a reduced generalized FPR disparity in general and violent recidivism prediction. In conclusion, Karimi et al. gave a future research direction that exploring different jurisdiction datasets of recidivism is essential to understand the importance of calibration further.

More on this note, we observed from our study that one of the significant focuses of scholars and researchers at the post-processing phases in fair predictive algorithms for predicting the risk of recidivism is towards ensuring equalized odds. However, it may be difficult to simultaneously hold equalized odds with one of these fairness criteria: demographic parity, accuracy parity, or predictive rate parity [65].

4 CONCLUSION AND FUTURE DIRECTION

This paper conducted a comprehensive review on fairness as a requirement of trustworthy AI when predicting the risk of recidivism. The different practical propositions and implementations in the literature for achieving fair AI models for predicting the risk of recidivism at the different AI development phases were analysed. Based on this analysis, we point out the following practical observations and future directions that need to be addressed to achieve further fair AI systems for predicting the risk of recidivism.

First, a step towards achieving fair AI systems for recidivism prediction is to assess the pre-processing, in-processing, and post-processing stages of AI development. However, unlike the pre-processing and post-processing phases, we observed that much practical work had not been done in the in-processing phase of trustworthy AI development for recidivism prediction. Figure 1 illustrates this imbalance. Therefore, a future direction is to focus more on the in-processing phase and make more practical propositions and implementations to achieve fair AI models for predicting the risk of recidivism.

Second, we observed that the different practical propositions and implementations to achieve fair AI models for predicting recidivism are targeted and applied at a desired and particular development phase without considering other phases of trustworthy AI development for recidivism prediction. This approach will not guarantee the achievement of fair AI systems predicting the risk of recidivism. Therefore, a future direction will be to practically implement the different unfairness mitigating techniques at each AI development phase, considering all three phases (pre-, in-, and post-processing) simultaneously.

Third, the pre-, in-, and post-processing phases are traditional approaches for accessing fairness in developing trustworthy AI. The Microsoft software teams developed a modern approach [2] that provides a robust methodological process for accessing and engineering the AI system development stages. The approach consists of 9 stages of the machine learning workflow. Therefore, a future direction will be to evaluate fairness through the ML workflow adopted by Microsoft software teams in developing fair AI models for predicting the risk of recidivism [2].

Moreover, the datasets used to train AI algorithms for predicting the risk of recidivism have not yet reached the optimization level required to achieve fair AI. Therefore, we must ensure good data quality and collate all necessary features for fair risk assessment systems. In addition, the COMPAS dataset, collated in Broward, USA, is the most widely and commonly used dataset for addressing and evaluating the fairness of AI systems for predicting the risk of recidivism. Therefore, it is dire to source datasets from other countries and jurisdictions to assess further fairness's impact on convicted individuals of a possible risk of recidivism.

Table 1: TN: True Negative, FP: False Positive, FN: False Negative, FPR: False Positive Rate, FNR: False Negative Rate, TNR: True Negative Rate, TPR: True Positive Rate, PPV: Positive Predictive Value, NPV: Negative Predictive Value, AUR-ROC: Area under the ROC Curve, MAE: Mean Absolute Error, RMSE: Root Mean Square Error, ANN: Artificial Neural Networks, LR: Logistic Regression, SVM: Support Vector Machines, CART: Classification and Regression Trees, RF: Random Forests, XGBoost: eXtreme Gradient Boosted, DT: Decision Trees, LDA: Linear Discriminant Analysis

Ref	Dataset	Algorithm Model Used	Fairness Pipeline	Proposition	Crime	Evaluation Metrics
[58]	Broward County, Florida dataset & Kentucky dataset	RiskSLIM, Explainable Boosting Machine	Post-processing	Employed the use of two fairness definitions: calibration and balanced group AUC (BG-AUC)	General, Violent, Drug, Property, Felony, Misdemeanor	AUC
[49]	City Attorney's case Management System	Regularized Logistic Regressions, Decision Trees, Random Forests, Extra Tree Classifiers	Post-processing	Methods for balancing equity	Misdemeanor	FPR, FDR, FOR, FNR, FP, FN, Recall
[38]	Juvenile Justice System of Catalonia	Logistic Regression, Support Vector Machine, K-Nearest Neighbors, Random Forest	Pre-processing, In-processing	Explored Equalizing base rates and Learning fair representations	Violence and Aggression	FPR, FNR, AUR-ROC
[29]	Florida Department of Corrections (FDOC) & Florida Department of Law Enforcement	K-Nearest Neighbours, Random Forests, AdaBoost, Artificial Neural Networks	Pre-processing	Proposed Singular Race Models to segment the dataset based on race	Violent, Property, Drugs, Public order	Accuracy, FPR, FNR
[27]	Recidivism of Prisoners Released in 1994	Artificial Neural Network	Pre-processing	Proposed a new fairness measure, Bias Parity (BP) score, to permit selection of the lowest bias model	All crimes	PPV, FPR, Statistical Parity, Treatment Equality, Bias Parity
[10]	Miami-Dade County Clerk of the Court	Multi-task Cascaded Convolutional Network (MTCNN)	Pre-processing	Employed using mugshots to help predict missing defendant race information in criminal courts datasets	Unspecified	Accuracy
[8]	Broward County, Florida (COMPAS)	Gradient Boosted Algorithm'	Pre-processing	Proposed a Combinatorial Algorithm for Proportional Equality (CAPE) to tackle prior probability shifts	Unspecified	FPR, FNR, Prevalence Difference (PD), Proportional Equality (PE)
[53]	Federal Probation System - Post Conviction Risk Assessment (2009-2019)	Post Conviction Risk Assessment (PCRA) algorithms	Pre-processing	Compared 'debiasing' algorithms and its effect on (1) calibration and positive predictive value & (2) False positive and false negative error rates tradeoff	Violent	AUC, PPV, FPR, FNR
[20]	Broward County, FL (COMPAS)	Neural Network (ADAM)	Pre-processing, Post-processing	Analyze the consequences of intersectionality and three novel fairness metrics satisfying intersectionality	Unspecified	Accuracy, Score, AUC-ROC
[14]	Broward County, Florida (COMPAS)	Logistic Regression, Random Forest	Pre-processing	Proposed a convex optimization for learning a data transformation error rates while preserving calibration	Felony, Misdemeanor	AUC-ROC
[28]	Recidivism of Prisoners Released in 1994	Artificial Neural Network	Pre-processing	Proposed an approach to select a model to increase recidivism prediction accuracy	Any crime, Sexual crime	Accuracy, FPR parity, FPR, FNR, TPR, TNR
[5]	Broward County, Florida (COMPAS)	Linear regression, Logistic Regression	In-processing	Introduced fairness regularizers for regression problems	Violent	Accuracy, MSE
[3]	Broward County, Florida (COMPAS)	Vanilla logistic regression	In-processing	Proposed penalizers aimed at reducing the difference between FPR and FNR	Unspecified	Accuracy, FPR, FNR
[48]	Broward County, Florida (COMPAS)	Logistic Regression, Support Vector Machine	Post-processing	Proposed a unified machine learning framework capable of handling any definitions of fairness	Unspecified	Accuracy, FPR, TPR
[9]	Broward County, Florida (COMPAS)	Logistic Regression	In-processing	Reformulate algorithmic fairness as constrained optimization with the objectives of maximizing public safety	Violent	Accuracy, AUC, FPR
[30]	RisCanvi Dataset	Logistic Regression, Multi-layer Perceptron, Support Vector Machine	Post-processing	Applied a relaxation method to satisfy equalized odds or parity in the error rates while preserving calibration	Violent, General	AUC, FPR, FNR
[65]	Broward County, Florida (COMPAS)	Logistic Regression, Neural Network	Pre-processing	Proposed a novel representation learning algorithms and an optimization formulation that demonstrates the importance of a Balanced Error Rate	Unspecified	
[39]	Broward County, Florida (COMPAS)	Random Forest	Post-processing	Developed a method to generate predictors that satisfy the fairness criterion approximate counterfactual equalized odds	Misdemeanor, Felony	FPR, FNR
[33]	Broward County, Florida (COMPAS)	Logistic Regression	Pre-processing, In-processing, Post-processing	Proposed a One-vs-One mitigation method to mitigate intersectional bias	Unspecified	Balanced Accuracy
[61]	Broward County, Florida (COMPAS)	Logistic Regression	Post-processing	Proposed a FairBayes-Difference in Predictive Parity (DPP), to estimate the fair Bayes-optimal classifier under predictive parity	Misdemeanor	Accuracy, FPR, FNR, Accuracy
[47]	Broward County, Florida (COMPAS)	Logistic Regression	Post-processing	Investigated the tension between minimizing error disparity across while maintaining calibrated probability estimates	Unspecified	FPR, FNR, Accuracy

ACKNOWLEDGMENTS

This work was supported in part by the Science Foundation Ireland grants 13/RC/2094_P2 (Lero) and 13/RC/2106_P2 (ADAPT) and is co-funded under the European Regional Development Fund (ERDF).

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